Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Phase I/II, Dose Finding and Optimization Study of [177Lu]Lu-NeoB in Combination With Capecitabine in Patients With GRPR+, ER+, HER2- Metastatic Breast Cancer After Progression on Previous Endocrine Therapy in Combination With a CDK4/6 Inhibitor **Short Title/Acronym:** NeoB-Cap1 **Protocol Number:** CAAA603D12101 **NCT Number:** NCT06247995 **Posting ID:** 390008215936574183 **Trial Status:** Recruiting **Sponsor Name:** Novartis Pharmaceuticals (Novartis) **Investigational Product:** [177Lu]Lu-NeoB (radioligand therapy) in combination with Capecitabine. Diagnostic imaging with [68Ga]Ga-NeoB **Phase of Development:** Phase I/II **Medical Monitor:** [Name and Contact Information]

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2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** * *Phase I (Dose Escalation):* To determine the recommended doses (RD) and dosing regimens of [177Lu]Lu-NeoB in combination with capecitabine based on dose-limiting toxicities and adverse events. * *Phase II (Dose Optimization):* To evaluate preliminary anti-tumor activity (including objective response rate, clinical benefit rate, progression-free survival, and overall survival) across two different doses/regimens. * **Secondary Objectives:** To evaluate pharmacokinetics (PK) and biodistribution (dosimetry) data of the therapeutic combination.

2.2 Background & Rationale To address the optimal sequence of therapy in patients with estrogen receptor-positive (ER+), human epidermal growth factor receptor-2 negative (HER2-) metastatic breast cancer (mBC) after disease progression on CDK4/6 inhibitors and endocrine therapy. [177Lu]Lu-NeoB is a first-in-class radioligand therapy that selectively binds to the Gastrin Releasing Peptide Receptor (GRPR), which is widely expressed in ER+ breast cancer. Capecitabine is a standard chemotherapy and a potent radiosensitizer; its synergistic combination with a radionuclide has the potential to overcome endocrine resistance and enhance treatment efficacy.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Dose-finding / Active evaluation **Blinding:** Open-label **Randomization:** Randomized (in the Phase II dose optimization segment to evaluate two higher dose levels)

3.2 Planned Enrollment & Duration Target N: Approximately 58 participants (between 36 and 58; approx. 18 in Phase I; 28-40 in Phase II) **Number of Sites:** Multicenter **Estimated Study Start:** [Early 2024] **Estimated Primary Completion:** [DD-MMM-YYYY]

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult men, and pre/peri/postmenopausal women (pre/peri-menopausal limited to Phase II). 2. Histologically and/or cytologically documented diagnosis of ER+ (>10% expression), HER2- metastatic breast cancer. 3. Radiologically confirmed progression of disease after 1 to 3 prior lines of endocrine therapy, one of which must have included a CDK4/6 inhibitor. 4. Target lesion showing adequate uptake of [68Ga]Ga-NeoB (indicating GRPR expression) on PET/CT or PET/MRI screening.

4.2 Exclusion Criteria 1. Pregnant or breastfeeding women. Highly effective contraception is strictly required. 2. History of or ongoing acute pancreatitis within 1 year of screening. 3. Impaired cardiac function, clinically significant cardiac disease, or ECG abnormalities indicating significant safety risks.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Investigational Products: * [177Lu]Lu-NeoB * Capecitabine (Trade Name: Xeloda | ATC Code: L01BC06) - *Reference: EMA Product Information*

Dosage/Frequency: * *Phase I:* [177Lu]Lu-NeoB at 150 mCi every 6 weeks (Q6W) + capecitabine (1000 mg/m2 twice daily for 14 days, followed by 7 days off). * *Phase II:* Randomized to higher dose regimens (e.g., 200 mCi Q6W or 100 mCi Q3W) + capecitabine. **Route of Administration:** Intravenous (IV) for [177Lu]Lu-NeoB and diagnostic agent; Oral for capecitabine. **Duration of Treatment:** Treatment cycles up to 18 months.

5.2 Concomitant & Prohibited Therapy Permitted: Gonadotropin-releasing hormone agonist (GnRha) for men and pre/peri-menopausal women. Standard supportive care. **Prohibited:** Hormonal methods of contraception (oral, injected, implanted, or transdermal). Use of other investigational drugs.

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6. Study Timeline & Visit Schedule

| Visit ID | Window | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Up to 6 weeks | Informed Consent, Medical History, PET/CT or PET/MRI with [68Ga]Ga-NeoB | | **Treatment Cycles** | Every 3 or 6 weeks | Study drug administration, vital

signs, safety labs | | **Tumor Assessments**| Every 9 to 12 weeks | Tumor evaluation (Q9W until Month 18, then Q12W until progression) | | **End of Treatment** | + 8 weeks | Final safety follow-up 8 weeks after last study treatment | | **Long-term Follow-up**| Up to 5 years | Post-treatment survival and long-term safety monitoring |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: * *Phase I:* Incidence and severity of dose-limiting toxicities (DLTs) and adverse events. * *Phase II:* Objective Response Rate (ORR), clinical benefit rate, progression-free survival (PFS), and overall survival (OS). **Measurement Method:** RECIST v1.1 criteria and CTCAE safety grading.

7.2 Secondary & Exploratory Endpoints * Pharmacokinetic (PK) parameters. * Biodistribution and dosimetry data.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection per National Cancer Institute (NCI) CTCAE guidelines. **Serious Adverse Events (SAEs):** Reported within standard 24-hour oncology protocol timelines. **Data Monitoring Committee (DMC):** Internal/External safety review committee for dose escalation decisions.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for Phase II efficacy; Safety Set for DLT and AE evaluations. **Sample Size Justification:** Approximately 58 participants; sized for dose-escalation toxicity probability (Phase I, n~18) and signal-seeking preliminary efficacy (Phase II, n=28-40). **Handling of Missing Data:** Standard oncology imputation; censored data models for progression-free and overall survival analyses.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical monitoring and centralized independent review for radiological tumor assessments. **Audit Trail:** Standard EDC/eCRF locking procedures and data clarification forms.

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11. Study Identifiers & Governance

Primary ID: CAAA603D12101 **Registry Number:** NCT06247995 **Posting ID:** 390008215936574183 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** NeoB-Cap1 **Official Regulatory Title:** A Phase I/II, Open-label, Multi-center Trial of [177Lu]Lu-NeoB in Combination With Capecitabine in Adult Patients With Gastrin Releasing Peptide Receptor Positive, Estrogen Receptor-positive, Human Epidermal Growth Factor Receptor-2 Negative Metastatic Breast Cancer After Progression on Previous Endocrine Therapy in Combination With a CDK4/6 Inhibitor

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12. Clinical Framework (Oncology Specific)

Disease Area: Breast Cancer **Primary Condition:** Metastatic Breast Cancer (mBC) **Preferred UMLS Name:** ER-positive, HER2-negative metastatic breast cancer **Terminology Codes:** * **MeSH Code:** D001943 * **SNOMED ID:** 254837009 **Diagnosis Threshold:** ER+/HER2- status, progression post CDK4/6i, and confirmed GRPR expression via diagnostic PET imaging. **Medical Methodology:** Targeted Radioligand Therapy + Chemotherapy **Optimization Target:** Recommended dose and regimen of [177Lu]Lu-NeoB combined with capecitabine.

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13. Study Procedures & Methodology

Management Pathway: Targeted Oncology Therapy Protocol **Education Protocol (HCP):** Radiopharmaceutical handling, safety protocols, and DLT monitoring instructions. **Education Protocol (Patient):** Contraception requirements, radiation safety practices, and oral chemotherapy adherence tracking. **Quality Audit Mechanism:** Centralized RECIST 1.1 tumor assessments and dosimetric evaluation.

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14. Observation & Follow-Up Schedule

Total Duration: Up to 5 years **Onsite Visit Cadence:** Every 3 or 6 weeks during the active 18-month treatment period **Remote Monitoring Frequency:** As dictated by standard post-treatment oncology follow-up **Checklist Review Interval:** Imaging assessments every 9 weeks (+/- 7 days) until Month 18, then every 12 weeks

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] |** ___ **|** ***[DD-MMM-YYYY] | | Lead Statistician | [Name] |*** _____ | [DD-MMM-YYYY] |